

Event Chain-Driven Communication Strategy Generation for News Videos

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Abstract

This study presents an intelligent planning system (News Video to Propagation Rules Strategy, NV2PRS). The system is based on event chain modeling to achieve automated generation of event-level communication strategies for video news. It consists of two modules: video style feature extraction and knowledge chain matching. First, a multimodal feature analysis engine is used to obtain text semantics, visual features, and communication features. Then, template-based knowledge chain matching is employed to realize the mapping between events and strategies. To optimize the system's practicality, a hierarchical architecture design is adopted, integrating a feature visualization interface and an end-to-end workflow for strategy generation.

CCS Concepts

- Computing methodologies → Modeling and simulation → Model development and analysis;
- Information systems → Information systems applications → Decision support systems.

Keywords

Communication strategy Generation, Event chain modeling, Multimodal feature analysis.

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1 Introduction

Current approaches to generating communication strategies for video news are constrained by manual experience-driven models, facing bottlenecks such as low efficiency in batch construction of multimodal features and insufficient capability for modeling dynamic event evolution [1]. Building on breakthroughs in multimodal representation learning [2][3] and event causal reasoning techniques [4], this study innovatively proposes the NV2PRS dual-driven architecture system. It establishes a mapping mechanism from structured representation of video events to communication strategies, achieving automated generation and enhanced dynamic adaptability.

As shown in Figure 1, the NV2PRS system is designed based on a dual-driven architecture. First, a multimodal feature analysis engine decomposes multimodal features from video streams. Then, communication strategies are generated through mapping relationships in the event chain matching mechanism module. The multimodal feature analysis engine employs a hierarchical architecture to extract five-dimensional features from textual, visual, and communication modalities, and constructs a temporal link feature database using temporal information. The event chain matching mechanism module uses pre-analyzed chain-style communication strategy templates in the pre-analysis layer to perform rule-based matching on feature data, completing the mapping and conversion between events and strategies. Compared with traditional manual analysis methods [5][6], the system improves the efficiency of strategy analysis and generation through automation. Compared with existing single-dimensional analysis tools [7][8], its innovative adoption of multimodal feature chain-driven rule matching enhances the logical consistency and convenience of strategy mapping.

2 System Architecture

The NV2PRS system architecture consists of two core modules: the multimodal feature analysis engine and the event chain matching mechanism.

2.1 Multimodal Feature Analysis Engine

To This engine is responsible for video deconstruction, multimodal representation extraction, and temporal link feature database construction. It includes three submodules:

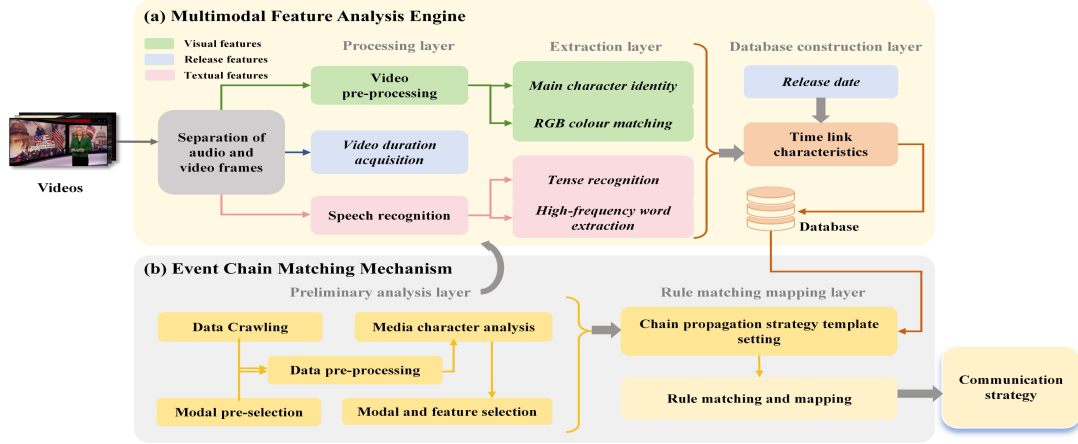


Figure 1: The pipeline of the proposed multimedia communication strategies generation system

2.1.1 Feature Processing Layer. To perform dual-stream preprocessing of video and audio. Comprising a multimodal data processor, it synchronously processes data streams from video decomposition [9]. The video stream undergoes frame rate extraction, frame segmentation, key frame extraction, and frame cropping via OpenCV [10], generating the visual feature intermediate object “key frame sequence” and extracting the propagation feature “video duration”. The audio stream generates the text feature intermediate object “transcribed text” through Whisper [11] speech transcription.

2.1.2 Feature Extraction Layer. To implement visual and text dual-modal feature extraction. Comprising a dual-modal feature extractor, it receives the key frame sequence and transcribed text from the feature processing layer. The visual feature extractor uses PaddleOCR [12] to identify subtitles containing subject identity information in key frames, extracting the visual feature “narrative subject” and obtains RGB [13] color channel values via OpenCV to analyze the visual feature “overall hue”. The text feature extractor performs tokenization, cleaning, word frequency statistics, and part-of-speech tagging via NLTK [14] to parse the text features “high-frequency words” and “overall tense”.

2.1.3 Database Construction Layer. To constructs the temporal link feature database. This layer integrates multimodal feature data based on temporal information, expands the five-dimensional features obtained from the previous two layers into six-dimensional features by incorporating the propagation feature “release date” and constructs the database accordingly.

2.2 Event Chain Matching Mechanism

This mechanism realizes the mapping between multimodal video representations and propagation strategies based on preset rule templates. It includes two submodules:

2.2.1 Pre-analysis Layer. Responsible for constructing rule templates for event chain propagation strategies. The construction logic is based on the analysis of hot topic video news from multiple mainstream media: First, news video sets across platforms with the same theme are crawled, and three analysis

perspectives—textual, visual, and propagation modalities – are preset. After multimodal data preprocessing, the representation rules of different modalities in each medium are parsed to screen a set of modal features with significant discrimination to support the multimodal feature analysis engine, while generating event chain rule mapping templates.

2.2.2 Rule Matching and Mapping Layer. To perform the mapping between temporal representation data and propagation strategies. Based on preset rule templates, this layer receives temporal representation data from the analysis engine. First, it performs temporal fractal segmentation on the temporal feature data according to temporal intervals and popularity fluctuations caused by indicators such as video views and shares, generating four-stage event chain feature clusters representing the propagation lifecycle: preheating, development, climax, and ending. Second, it conducts cluster-level multimodal rule statistics and strategy mapping according to the matching templates, covering:

2.2.2.1 Propagation Dimension: Propagation duration, ratio of long/short video releases, and release strategies.

2.2.2.2 Visual Dimension: Narrative subject, subject release strategy, and high-frequency color tones.

2.2.2.3 Textual Dimension: Dominant tense and core high-frequency word sets.

Finally, the mapped content of each cluster is integrated through event chains to realize a complete mapping from multimodal features to propagation strategies.

3 User Interface

The demonstration video showcases the user interface design of the NV2PRS system, which adopts a hierarchical visualization panel architecture. The design follows the system’s dual-module workflow to enable low-threshold interaction throughout the entire process from raw video input to end-to-end strategy output, allowing users to quickly generate multimedia communication strategies without complex operations.

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